

Problem Statement: Loan Approval Prediction using Machine Learning

Financial institutions receive a large number of loan applications every day from individuals with diverse financial and personal backgrounds. Evaluating each application manually is a time-consuming and error-prone process that may lead to inconsistent decisions, delayed approvals, and increased operational costs. Incorrect loan approval decisions can result in financial losses for banks, while rejecting eligible applicants may reduce customer satisfaction. The objective of this project is to design, build, and evaluate a **binary classification model** that accurately predicts whether a loan application should be **approved or rejected** based on applicant information such as gender, marital status, education, employment status, income, loan amount, loan term, credit history, and property area. The goal is to develop a reliable machine learning solution using **IBM watsonx.ai AutoAI** that assists financial institutions in making faster, more accurate, consistent, and data-driven loan approval decisions.